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Later bells, longer sleep

A cluster-randomised trial of school start times in 24 secondary schools

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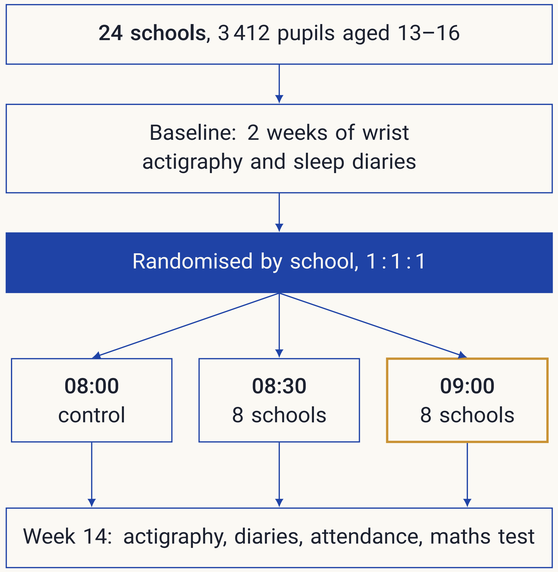
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1 Background

Adolescents fall asleep about two hours later than children [1], yet most schools start before 08:30. The result is chronic short sleep on school nights [2].

Observational studies link later starts to longer sleep [3, 4], but randomised evidence is scarce [5].

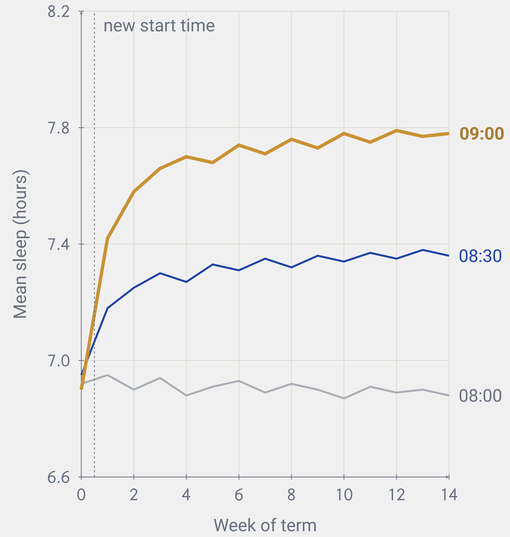
2 Study design



Schools, not pupils, were randomised, so timetables changed for everyone. Analysis by intention to treat with a random intercept per school.

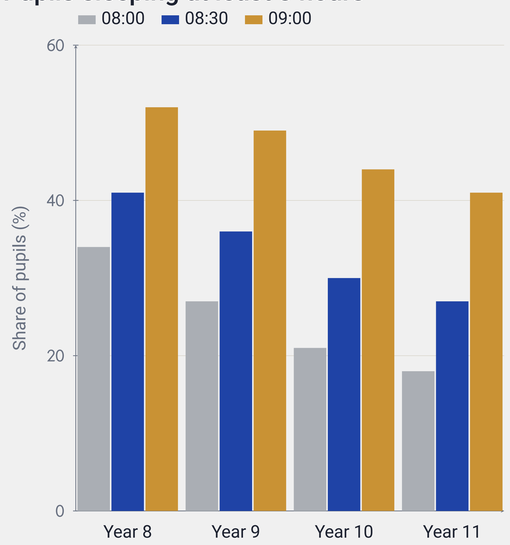
3 Results

Sleep on school nights



Weekly means from actigraphy, all pupils. The 09:00 arm gains most of its extra sleep within three weeks; by week 14 the gap is 54 minutes.

Pupils sleeping at least 8 hours



Week 14, school nights. The gap grows with age, when the body clock runs latest.

Secondary outcomes at week 14, 09:00 against 08:00

Outcome	08:00	09:00	Difference	95% CI
Sleep onset (clock time)	23:21	23:28	+7 min	−2 to +16
Wake time (clock time)	06:38	07:39	+61 min	+52 to +70
Daytime sleepiness, PDSS	17.8	14.6	−3.2	−4.1 to −2.3
Attendance (%)	92.1	94.0	+1.9	+0.8 to +3.0
Maths test (*z*-score)	0.00	0.11	+0.11	+0.02 to +0.20

Mixed-effects estimates adjusted for baseline value, year group and sex; $n = 3\,187$ pupils with follow-up data.

Headline

+54

minutes of sleep per school night at 09:00, against 08:00. Sleep onset barely moved, so the gain is almost all wake time.

4 Take-home messages

1. A 09:00 start gives teenagers almost an hour more sleep per night.
2. The gain appears in the first three weeks and holds for the whole term.
3. Pupils are less sleepy, attend more and score slightly higher in maths.

5 Limits and next steps

Pupils and teachers knew the start time, so diaries may be biased; actigraphy is not.

One term cannot show lasting effects. A two-year follow-up of all 24 schools reports in 2028.

6 Acknowledgements

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FULL PAPER AND DATA
https://example.org/later-bells